



Pharm*Achieve*

HEADACHES AND MIGRAINES

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

5-HT Serotonin

A/E Adverse Effect

ASA Acetylsalicylic Acid

CBT Cognitive Behavioural Therapy

C/I Contraindication

CGRP Calcitonin Gene-Related Pep

DHE Dihydroergotamine

HA Headache

HTN Hypertension

MAOI Monoamine Oxidase Inhibitor

MOH Medication Overuse Headache

NSAID Nonsteroidal Anti-Inflammatory Drug

OC Oral Contraceptive

QOL Quality of Life

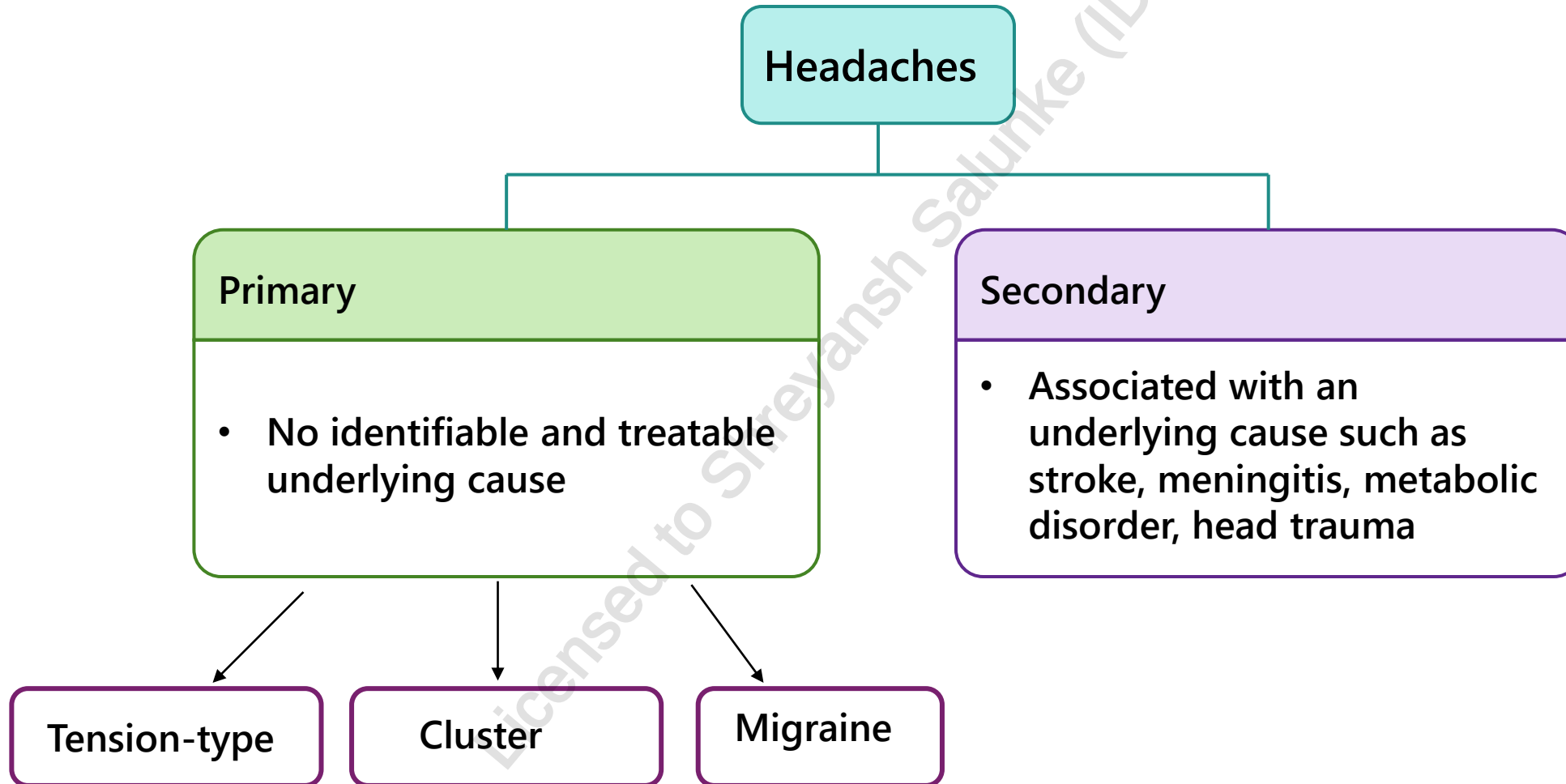
SNRI Serotonin-Norepinephrine Reuptake Inhibitor

SSRI Selective Serotonin Reuptake Inhibitor

TCA Tricyclic Antidepressant

TTH Tension-type Headache

TYPES OF HEADACHES



IMPORTANT QUESTIONS TO ASK

Onset of headache

Pain location

Headache-associated symptoms

Precipitating factors

Remitting factors (e.g. drug therapy)

Presence of comorbidities

RED FLAGS FOR SECONDARY HEADACHES - SNOOPPPP

Systemic: symptoms of stiff neck, fevers, chills, night sweats, weight loss, jaw pain, history of malignancy, immunosuppression or HIV

Neurologic: diplopia, motor weakness, pulsatile tinnitus, eye pain (uni or bilateral) with diminished vision, behaviour or personality changes

Onset: sudden, peak onset within 1 minute and lasts for 5 or more minutes (thunderclap), access for stroke, subarachnoid hemorrhage

Older age: >50 years old

Pattern change: postural headache (symptoms worsen when standing up), change in headache characteristics or progressive headache

Papilledema: swelling of the optic discs in the eyes determined through physical examination

Precipitated: by coughing, sneezing, exercise or Valsalva maneuver

Pregnancy: new onset headache or change in headache during pregnancy



TYPES OF HEADACHES

Tension Type Headache	Cluster Headache	Migraine
Mild, dull constant aching, pressure	Severe attack of piercing pain	Pulsing, throbbing (may start with TTH) Patients often complain of a recurring "sinus" headache
Bilateral	Unilateral	
Bilateral dull aching sensation on the top of the head, "tight band"	Behind the eye or over the lateral aspect of nose	Temporal, eye region (lower than forehead)
Radiates to neck and back; Not accompanied by GI symptoms*	Accompanied with nasal secretions, lacrimation	Accompanied with GI symptoms* and/or light sensitivity
No aura	Possible aura	Aura ~10-30 minutes before HA (should not last > 1 hr)
Not triggered by foods or activity; precipitated by stress	Precipitated by alcohol use, changes in season, naps	Triggered by foods (cheese, chocolate), fatigue, stress; aggravated by activity
May last for hours-days (up to 7 days)	Duration: 15 mins - 3 hours [several times/day] Sudden onset (3-5 min peak)	Duration: 4-72 hours
At least 2 of: bilateral headache, non-pulsating pain, mild-moderate pain, not worsened by activity	Cluster period may last several weeks to months	At least 2 of: nausea, light sensitivity, interference with activities
*No nausea or vomiting - Can't have both photophobia or phonophobia		*Presence of nausea or vomiting and/or - Photophobia, phonophobia, osmophobia

DETERMINE THE MOST APPROPRIATE DIAGNOSIS



"I feel pressure around my head; it's a dull, constant aching."



"I feel piercing pain behind my left eye."



"I feel light sensitivity and nausea"

Presentation:

RISK FACTORS

TTH

- Stress
- Excessive caffeine
- Excessive alcohol
- Lack of sleep
- Poor nutrition/dehydration
- Poor posture
- Odours
- Smoke
- Sunlight

VS

MIGRAINE

- Stress
- Caffeine
- Alcohol
- Fatigue
- Fasting
- Genetics
- Menses
- Changes in atmospheric pressure
- Altitude changes
- Trigger foods (changes per individual)

MEDICATION-OVERUSE HEADACHE

ANALGESIC-INDUCED/REBOUND

- HA occurring at ≥ 15 days per month for ≥ 3 months
- Leads to a cycle where headache returns as soon as effects of analgesics wear off, causing the patient to use more medication for relief
- Occurs with:
 - Simple analgesics (acetaminophen, NSAIDs) used ≥ 15 days/month for ≥ 3 months
 - Opioids, triptans, analgesic-opioid combinations used ≥ 10 days/month for ≥ 3 months
- Symptoms present as TTH
- Recommended treatment:
 - Gradual discontinuation of opioid-containing agents
 - Abrupt or gradual discontinuation with simple analgesics/triptans
- May need to start prophylactic medications
- Manage comorbidities that may be contributing to headaches, such as depression

SUBSTANCE-INDUCED CAUSES (OTHER THAN MOH)

- Alcohol
- Cocaine
- Methylphenidate
- Caffeine withdrawal
- Corticosteroids
- Nitrates (nitroglycerin) and nitrate-containing deli meats
- SSRIs
- Oral contraceptives



Drugs associated with intracranial HTN (leading to HA):

- Tetracycline antibiotics
- Isotretinoin
- Tamoxifen

ORAL CONTRACEPTIVES

- Precipitate migraine attacks in females with no history of migraines
- Onset: within first few months-years of OC use
- After OC discontinuation, patients see improvement in HA symptoms within few months
- Cause: estrogen content
- Clinical pearls:
 - Estrogen-containing OCs are contraindicated in women with migraine history with aura at any age due to increased stroke risk
 - OCs may be considered in patients who have migraine history without aura (discuss risks vs benefits with healthcare provider)

GOALS OF THERAPY

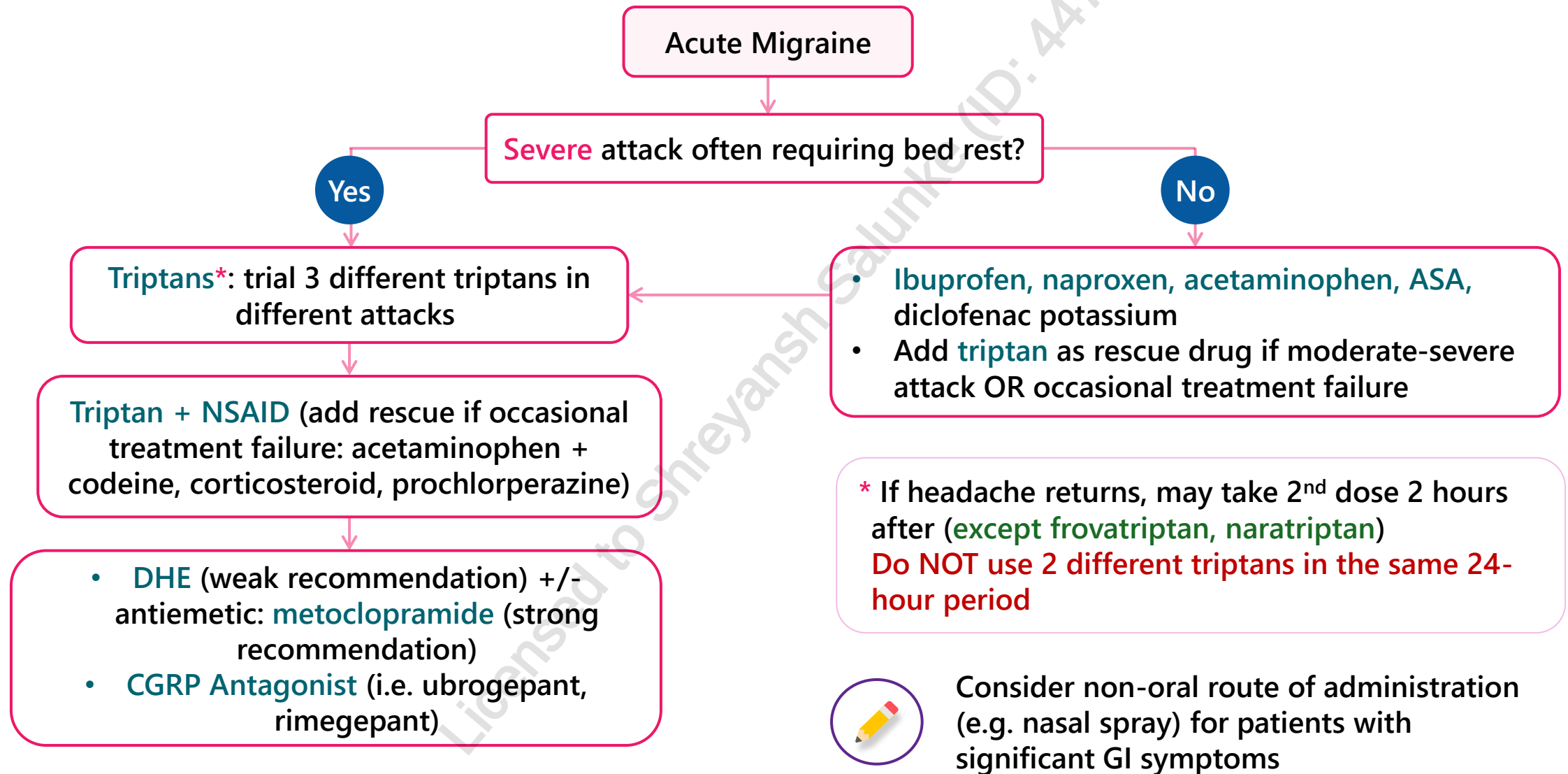
- ✓ Identify red flags and those requiring immediate referral
- ✓ Prevent medication-overuse headaches
- ✓ Identify drug-induced/reversible causes
- ✓ Relieve pain and associated symptoms with drug and non-drug strategies
- ✓ Reduce frequency, severity, duration, and disability of attacks

NON-PHARMACOLOGICAL MEASURES

- **Headache diary** → trigger management/avoidance if possible
- Apply **ice** and **rest** in a dark, noise-free room (heat may help for TTH)
- **Stress Management**: CBT, relaxation therapy, psychotherapy
- Acupuncture and/or nerve blocks
- **Routine**: healthy diet, regular exercise, good sleep hygiene
- Good **posture**, ergonomics at work (TTH)

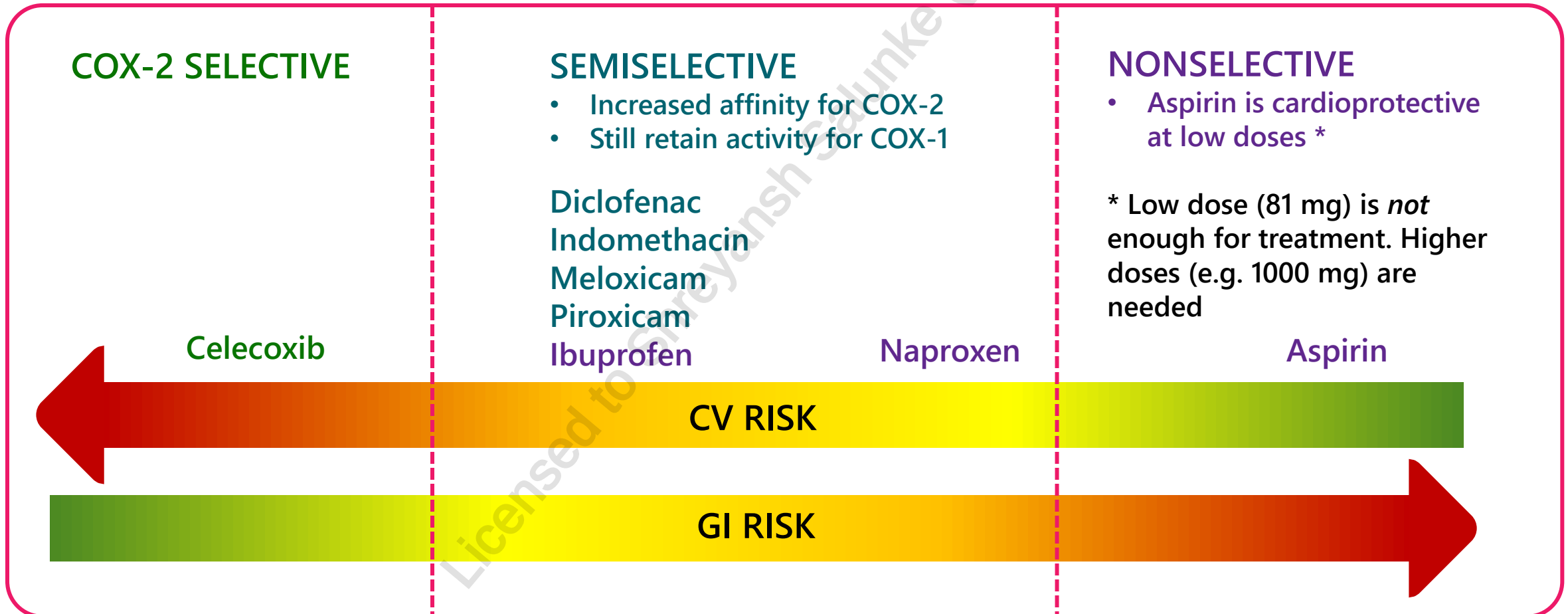
Determine patient's expectations and explain management options

ACUTE MIGRAINES TREATMENT



NSAIDs

- 1st line for mild-moderate migraines (not requiring bedrest) or as adjunct with triptan
- 1st line for tension-type headaches



TRIPTANS

- **First line for migraines**
- **Mechanism:** acts on serotonin (5-HT) subclass 1B and 1D receptors found on blood vessels and neurons to inhibit the release of vasoactive neuropeptides and cause vasoconstriction of the pain-sensitive blood vessels
- **Administration:** Take at first sign of migraine, repeat dose after 2 hours if partial relief (except frovatriptan, naratriptan)
 - Onset: 30-60 min
 - 2nd triptan dose unlikely to be effective if 1st dose provided no relief
 - **Do not use a different triptan within 24 hours of another triptan**
- **Use <10 days/month** to avoid medication overuse headache
- **Different formulations:**
 - **Oral wafer:** rizatriptan 10 mg or zolmitriptan 2.5 mg **if fluid ingestion worsens nausea**
 - **Nasal spray:** zolmitriptan 5 mg or sumatriptan 20 mg **if patient is nauseated**
 - **Subcutaneous:** sumatriptan 6 mg **if vomiting early in the attack/resistant to oral triptans**

TRIPTANS

- A/E: chest discomfort, fatigue, dizziness, paresthesia, drowsiness, nausea, throat irritation
 - **Caution**
 - Serotonergic drugs (SSRIs, SNRIs, MAOIs, ergots, triptans) → increased risk of serotonin syndrome
 - Drugs inducing and inhibiting CYP enzymes
 - **C/I:**
 - Heart disease
 - Uncontrolled HTN
 - *Pregnancy
 - Basilar or hemiplegic migraine
- * Research suggests sumatriptan safe during pregnancy; insufficient research to confirm safety of other triptans, however, evidence is reassuring. Avoid post-partum due to possible increase of stroke/angiopathy risk

TRIPTANS



Avoid taking an ergot derivative within 24 hours of a triptan!

Drug	Earliest repeat dose	Metabolism	Comments
Sumatriptan	2h (2 doses/24h)	MAO _A	• PO, SC, nasal formulations • PO has clear dose-response relationship (100 mg - sig better relief at 2h) • Onset of Action: tablets (30-60 minutes), nasal/SC (approx. 10 minutes) • Use SC if patient is vomiting early in the attack • Do NOT use with MAOIs • Available in combination with naproxen sodium (Suvexx®)
Zolmitriptan	2h (2 doses/24h)	CYP 1A2, MAO _A , renal	• Orally dispersible tablets can be taken without water • Renal clearance reduced with severe impairment • Nasal spray had faster onset of action and greater efficacy versus 2.5 mg tablet • Do NOT use with MAOIs
Frovatriptan	4-24h (2 doses/24h)	CYP1A2	• OCs and propranolol may increase serum concentrations by 30-60%
Naratriptan	4h (2 doses/24h)	CYP 3A4, 1A2, renal	• Slow onset, low incidence of side effects, overall rate of AE similar to placebo • 2.5 mg dose significantly more effective vs. 1 mg dose
Rizatriptan	2h (2 doses/24h)	MAO _A	• Use with caution in patients taking propranolol (increased bioavailability of rizatriptan) • Fast-melt wafers (i.e. for those who have worsening nausea with fluid ingestion), can be taken without water, may have rapid onset • Do not use with MAOIs
Almotriptan	2h (2 doses/24h)	CYP 3A4, 1A2, MAO _A , renal	• Do not use with MAOIs • Inhibitors of CYP3A4 may increase bioavailability
Eletriptan	2h (2 doses/24h)	CYP 3A4, 1A2	• Contraindicated within 72 hours of potent CYP3A4 inhibitors

ERGOT DERIVATIVES (DIHYDROERGOTAMINE/DHE)

FYI

- **Mechanism:** similar actions to the triptans but also interacts centrally with dopamine and adrenergic receptors
- **Route of administration:** injection or intranasal
 - Intranasal: convenient, bypasses the GI tract
- **Not used routinely** for acute migraine attacks as inferior to triptans and more side effects
- NO risk of dependence
- Use is limited by AEs: N/V (may require anti-emetics)
 - Normally preceded by metoclopramide to prevent DHE-associated nausea
 - Ergotism - vasoconstriction resulting in cold and painful extremities/tingling, muscle pain, chest tightness, coronary vasospasm
 - Intranasal DHE is associated with mild-moderate rhinitis
- **C/I: CVD, PVD, sepsis, liver/kidney disease, pregnancy**
- **Drug interactions:** Contraindicated in patients taking potent inhibitors of CYP3A4

ANTIEMETICS

- Medications to target nausea and vomiting

Anti-nauseant

Dimenhydrinate

- Side effects:
 - Sedation
 - Anticholinergic effects
 - Confusion

Antiemetic/prokinetic

Metoclopramide

- Side effects: diarrhea, cramping, headache, drowsiness, extrapyramidal symptoms, tardive dyskinesia

Domperidone

- Side effects: diarrhea, cramping, headache, QTc prolongation

CALCITONIN GENE-RELATED PEPTIDE (CGRP) INHIBITORS

- Newest class of migraine medications, **block the action of CGRP = stops/prevents migraine**
- **CGRP facilitates migraines** through various mechanisms (i.e., arterial vasodilation, inflammation in the dura, facilitation of neuronal signaling and modulation of nociceptive signal transmission at the trigeminal cervical complex)
- Side effects: nausea, dizziness, dry mouth, fatigue, injection site reactions (monoclonal antibodies)
- Should be avoided in pregnancy due to limited safety data (CGRP is important for blood flow regulation)

Preventative Treatment	Administration
Aimovig® (Erenumab)	Injection SC monthly
Ajovy® (Fremanezumab)	Injection SC monthly or once every 3 months
Emgality® (Galcanezumab)	Injection SC monthly
Vyepti® (eptinezumab)	IV once every 3 months
Qulipta® (atogepant)	PO tablet daily

Acute Treatment CGRP Antagonists	Administration
Nurtec ODT® (Rimegepant)	Oral disintegrating tablet prn
Ubrelvy® (Ubrogepant)	Oral tablet prn

CGRP BLOCKING AGENTS

Drug	Notes
Aimovig® Erenumab	- Caution: severe constipation, active or recent cardiovascular, cerebrovascular or peripheral vascular disease, patients planning pregnancy within 6 months and breastfeeding, - Side effects: injection site reaction, alopecia, muscle cramps, anaphylaxis, constipation
Qulipta® Atogepant	- Caution: severe constipation, kidney or liver disease, pregnancy and breastfeeding - Side effects: weight loss, drowsiness, hypertension
Vyepti® Eptinezumab	- Caution: same as erenumab - Side effects: nasopharyngitis, nausea, anaphylaxis, hypertension
Ajovy® Fremazumab	- Caution: same as erenumab - Side effects: same as erenumab
Emgality® Galcenezumab	
Nurtec ODT® Rimegepant	- Caution: in kidney or liver disease, pregnancy and breastfeeding - Side effects: drowsiness, hypertension, dizziness

MIGRAINE PREVENTION

Patient eligibility criteria:

1. ≥ 4 moderate–severe headache days a month NOT responding to acute medications
2. ≥ 8 headache days/ month, even when acute medications are effective
→ These patients are at an increased risk of medication-overuse headaches
3. Migraine attacks with significant impact on the patient's life despite the use of acute medications, lifestyle modification and trigger management

Considerations when choosing preventative therapy:

- Drug side effect profile
- Comorbid disorders (e.g. depression, hypertension, obesity, etc.)
- Planning pregnancy, pregnant or breastfeeding
- Patient choice in administration

Duration of therapy

- Therapy of choice should be trialled at the target dose for ≥ 2 months
 - OnabotulinumtoxinA should be injected at least twice
- Successful prophylaxis is identified by $\geq 50\%$ reduction in headache frequency or intensity
- No clear guidance on when to discontinue therapy, refer to specialist

2024 MIGRAINE PREVENTION GUIDELINE RECOMMENDATIONS

Drug	Indication
Aimovig® Erenumab	• Episodic migraines, strong recommendation & high-quality evidence • Chronic migraines, strong recommendation & high-quality evidence
Qulipta® Atogepant	• Episodic migraines, strong recommendation & moderate quality evidence • Chronic migraines, strong recommendation & high-quality evidence
Vyepti® Eptinezumab	
Ajovy® Fremazezumab	
Emgality® Galcenezumab	
Botox OnabotulinumtoxinA	• Chronic migraines, strong recommendation & high-quality evidence • NOT recommended for episodic migraines
Propanolol	• Chronic migraines, strong recommendation & moderate quality evidence
Candesartan	• Episodic migraines, strong recommendation & moderate quality evidence

MIGRAINE PROPHYLAXIS

✓ Triptans (frovatriptan, naratriptan and zolmitriptan) can be used prophylactically for menstrual-associated migraines

Patient Factors	Agent of Choice	Notes
Comorbid depression/ anxiety disorder	Amitriptyline Venlafaxine	Amitriptyline: anti-cholinergic side effects, caution concurrent use with serotonergic medications, cardiac disease Venlafaxine: caution use in patient at risk for long QT, side effects include insomnia, dry mouth, nausea, sweating
Overweight/obese	Topiramate Atogepant	Topiramate: downgraded to weak recommendation for episodic migraines, caution in patients planning pregnancy (teratogenic) or at risk of kidney stones
Comorbid hypertension	Beta-blockers Candesartan	Propranolol, metoprolol, nadolol - side effects include hypotension, dizziness, fatigue, and exercise intolerance. Caution beta-blocker use in patients with asthma, diabetes, and bradycardia Candesartan: hypotension, dizziness, caution in acute kidney injury or planning pregnancy
Pregnancy and Breastfeeding	Propranolol or metoprolol	Best to avoid migraine drug prophylaxis if possible. Recommended strategies include trigger avoidance and lifestyle modification. If drug prophylaxis is necessary, then propranolol or metoprolol are the best choice; if contraindicated or intolerable, then amitriptyline can be considered
Medication averse	Natural health products	Magnesium citrate, riboflavin (Vitamin B2) and coenzyme Q10
Treatment resistant migraines (failed properly dosed medication from ≥ 2 drug classes)	Anti-CGRP mABs Atogepant	Referral Consider combination therapy with Onabotulinum toxin A for chronic migraine (≥ 8 migraine days/ month and ≥ 15 headache days)

MIGRAINE PROPHYLAXIS

Patient Factors	Agent of Choice	Notes
Insomnia	Melatonin	Now a weak recommendation for use in episodic migraines
Epilepsy	Levetiracetam Topiramate Valporic acid	Levetiracetam: side effects include dizziness, drowsiness, and changes in mood or behaviour. Cautious use in patients at risk for depression or aggressive behaviour Valproic acid: side effects include tremors, fatigue, weight gain, hair thinning, Parkinsonism with long-term use, hepatic and pancreatic toxicity. Caution in patients with liver disease
Specific patient recommendations	Memantine Pizotifen Flunarizine Verapamil	Memantine: weak recommendation for episodic migraines. Caution use in patients with ongoing depression or psychiatric disease Pizotifen (serotonergic antagonist: weak recommendation. Caution use due to significant drug interactions. Anti-cholinergic side effects Flunarizine: weak recommendation. Caution use in patients with a history of depression or ongoing Parkinsonism Verapamil: weak recommendation. Caution use in patients with bradycardia, arrhythmia, and concurrent beta blockers. Side effects include hypotension, lower extremity edema, constipation

TENSION-TYPE HEADACHES

TREATMENT AND PROPHYLAXIS

Acute Treatment

- Non-pharmacological treatment (e.g. exercise, physiotherapy, massage therapy, acupuncture)
- Acetaminophen or NSAIDs (e.g. ibuprofen, naproxen)
- If frequency ≥ 15 days/month $\rightarrow$ chronic $\rightarrow$ refer for prophylaxis

Prophylaxis

- First-line: amitriptyline, nortriptyline
- Second-line: mirtazapine, venlafaxine

CLUSTER HEADACHES

TREATMENT AND PROPHYLAXIS

❑ Consider specialist referral

Acute Treatment

- Subcutaneous sumatriptan or intranasal zolmitriptan
- 100% Oxygen: 12 L/min for 15 min through non-rebreathing mask

Note: NSAIDs and opioids are not effective

Transitional prophylaxis (>2 attacks/day)

- Oral prednisone
- Occipital nerve blocks with steroids

Note: temporary bridge therapy while verapamil therapeutic levels are being achieved

Chronic prophylaxis

1. Verapamil
2. Lithium
3. Topiramate or melatonin

SPECIAL POPULATIONS

PREGNANCY

- **First Line: Non-pharmacological therapy** (e.g. relaxation strategies, hydration)



Migraine & tension-type headache abortive therapy

1. Acetaminophen
2. NSAIDs – ibuprofen/naproxen (**avoid if possible, especially in first and third trimester**)
3. Lidocaine injection (for severe migraines)

Nausea and/or vomiting

- Metoclopramide
- Prochlorperazine

Prophylaxis

- Propranolol or metoprolol (may consider discontinuing a few days before delivery and monitor neonates)

- **Triptans generally avoided** (growing evidence of safety for sumatriptan, but increased risk of hypertension and bleeding. **Consider after risk/benefit discussion with provider**)
- **Ergots are contraindicated**

SPECIAL POPULATIONS

BREASTFEEDING

- First Line: Non-pharmacological therapy

Abortive therapy

- Acetaminophen (preferred agent)
- Ibuprofen (avoid if possible, but can use)
- Sumatriptan (has been studied in lactation and is considered compatible, however, it is crucial to avoid vasoconstricting agents in the initial postpartum period)
 - Holding breastfeeding for 8-12 hours after dose will reduce exposure to infant

- Avoid ergot derivatives, barbiturates and opioids, acetylsalicylic acid



lactation may have a positive effect on migraine activity!

Prophylaxis

- Propranolol
- Metoprolol

SPECIAL POPULATIONS

CHILDREN

✓ Non-drug measures are always first-line!

Tension Type Headache

Abortive therapy:

First line: acetaminophen and NSAIDs (ibuprofen, naproxen)

Preventative therapy:

Amitriptyline, nortriptyline
Other options: venlafaxine, mirtazapine (less efficacy)

Migraine Headache

Abortive therapy:

First Line: ibuprofen, naproxen, acetaminophen; antiemetics as needed (metoclopramide, prochlorperazine, ondansetron)
Triptans: only almotriptan is Health Canada approved for use in patients 12 years of age and above; off-label use with others (evidence for oral almotriptan, nasal zolmitriptan/ sumatriptan in adolescents; oral rizatriptan in children as young as 6 years)

Preventative therapy:

Propranolol, topiramate, amitriptyline



Acetylsalicylic acid should be avoided in children <18 years of age as there is a risk of Reye syndrome

MONITORING PARAMETERS

EFFICACY ENDPOINTS

Parameter	Degree of Change	Timeframe
# of episodes/month	Reduce/eliminate	Variable depending on baseline frequency
Duration of episodes		
Quality of life	Improvement	Ongoing
Use of NSAIDs and acetaminophen	< 15 days/month	Throughout therapy
Use of opioids and triptans	< 10 days/month	

SAFETY ENDPOINTS

Parameter	Degree of Change	Timeframe
AEs (e.g. drowsiness, nausea, fatigue)	Prevent or eliminate	Throughout therapy
Serotonin Syndrome (e.g. agitation, tremor, sweating, tachycardia)	Prevent	

TAKEAWAY

- **Migraine:** Pulsing, throbbing (may start with TTH), patients will often complain of a recurring “sinus” headache. First-line for mild-moderate is NSAIDs. First-line for severe (requiring bed rest) is triptans
 - For prophylaxis: consider beta-blockers or TCAs
- **Tension type headache:** Mild, dull constant aching, pressure. First-line treatment is acetaminophen or NSAIDs
- **Cluster headache:** Severe attack of piercing pain. First-line is injectable triptan and oxygen
- **Medication overuse headache:**
 - HA occurring when simple analgesics (acetaminophen, NSAIDs) used ≥ 15 days/month
 - HA occurring when opioids, triptans, analgesic-opioid combinations used ≥ 10 days/month
- **Triptans Clinical Pearls:**
 - If partial improvement with triptan, can take 2nd dose after 2 hours (except frovatriptan, naratriptan)
 - May add NSAID with triptan or as rescue therapy after triptan failure
 - Try 3 different triptans in different headache episodes before deeming drug class ineffective
 - **Caution with triptan use and serotonergic drugs (SSRIs, SNRIs, MAOIs, ergots, triptans) due to increased risk of serotonin syndrome**
 - **Do not use in patients with cardiac-like symptoms, heart disease, uncontrolled hypertension or pregnancy. Do not use a triptan within 24 hours of another triptan**

CASE

MH is a 25-year-old female. She has been complaining of a throbbing headache on one side for the last day or so. She has also been feeling a little nauseous. Her past medical history is insignificant and the only medication she takes is melatonin 3 mg PO QHS PRN (1-2 times per month).

1. Based on MH's symptoms, what type of HA is she likely to have?

- a) Cluster
- b) Migraine
- c) Tension-type
- d) Thunderclap



CASE

MH is a 25-year-old female. She has been complaining of a throbbing headache on one side for the last day or so. She has also been feeling a little nauseous. Her past medical history is insignificant and the only medication she takes is melatonin 3 mg PO QHS PRN (1-2 times per month).

2. What would be a reasonable first-line option for her at this point?

- a) Naproxen in combination with a triptan
- b) Oral sumatriptan 100 mg
- c) Acetaminophen 500-1000 mg
- d) Metoclopramide 10 mg



CASE

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3. Assuming MH explains that her HA specifically occur around the time of her menstrual cycle and she is looking for something prophylactic, which of the following would be a first-line recommendation?
- a) Venlafaxine
 - b) Candesartan
 - c) Frovatriptan
 - d) Gabapentin



CASE

MH is a 25-year-old female. She has been complaining of a throbbing headache on one side for the last day or so. She has also been feeling a little nauseous. Her past medical history is insignificant and the only medication she takes is melatonin 3 mg PO QHS PRN (1-2 times per month).

4. Assuming MH returns with the same symptoms 2 years later, but is now pregnant, which of the following would be the most appropriate first-line agent for abortive therapy?
- a) Acetaminophen
 - b) Naproxen
 - c) Propranolol
 - d) Triptan



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CHANGE LOG

July 2025

- Formatting edits made throughout and slides re-arranged
- Slide 6: Modified red flags to fit into a mnemonic
- Slide 10: Added duration to MOH definition
- Added new slides 23,24 and 25
- Slides 26 and 27: Modified and simplified information
- Slide 28: Added more non-pharmacological interventions
- Slide 29: Modified information as per guidelines
- Slide 30: Added lidocaine injection for severe migraine
- Slide 38: Reference list updated

September 2025

- Slide 34-35: Added monitoring parameters slides

November 2025

- Slide 7: Addition of length of migraine aura
- Slide 15: Addition example of CGRP receptor antagonists
- Slide 19: Addition of Ergot DDI tip
- Slide 33: Consolidated monitoring parameters

January 2026

- Slide 16: Added note on the use of low dose ASA

April 2026

- Formatting edits made throughout
- Slide 7: Addition of osmophobia for migraines, addition of cluster headache onset and expanded duration
- Slide 14: Addition of TTH heat therapy
- Slide 22: Side effects and pregnancy note added
- Slide 32: Changed ASA use age limit to < 18 years old